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Distributed Motion Coordination Algorithms

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Overview

The objective is to study and simulate various algorithms for controlling multiple robots and to apply some of these to a group of wheeled robots. These algorithms include:

1. Algorithms to make a single robot move in a circle with specific centre and radius
2. First-order consensus [1]
3. Second-order consensus enabling a group of robots to drive to a goal (goal seeking) [2]
4. An algorithm for goal seeking, whilst maintaining formation and avoiding stationary obstacles [3]

Applications

Logistics, autonomous vehicles, agriculture

Consensus

- About the ability to agree on a value.
- Fundamental to many distributed algorithms, including algorithms A2-A4.
- First-order consensus protocols can minimise a function of the positions of a group of robots.
- Second-order consensus protocols can control position and velocity.

Results

A1. Circular Control

Proposed polar control law:

$$\mathbf{u} = \hat{\mathbf{r}}(-(r - r_d) - \dot{r}) - \hat{\boldsymbol{\theta}}(\dot{\theta} - \dot{\theta}_d)$$

where $\hat{\mathbf{r}}$ and $\hat{\boldsymbol{\theta}}$ are radial and tangential unit vectors in *cartesian coordinates* relative to the centre.

- The radius, radial velocity and angular velocity converge to r_d,0 and $\dot{\theta}_d$, respectively.
- Has no steady-state error and simpler than proposed cartesian control laws.

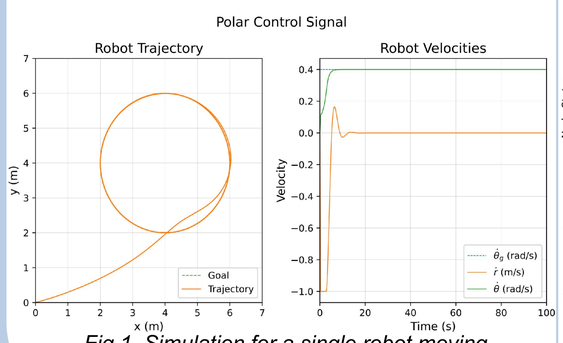


Fig 1. Simulation for a single robot moving in a circle using the control law above

A2. First-order Consensus

The simulations in [1] were verified. E.g. fig. 2 shows that several nodes can achieve consensus (under certain conditions) on their average despite the communication graph changing.

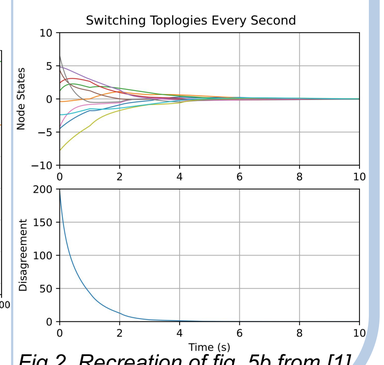


Fig 2. Recreation of fig. 5b from [1]

Off-axis Control

When converting the desired velocity in cartesian coordinates of the axis of a unicycle robot to a control velocity, (v, ω), the angular velocity is undefined. I.e. if a robot's velocity in the x and y directions is zero, it can still rotate on the spot. The following procedure is computed at each time step for the results below:

1. For each robot, get the current control velocity and heading, θ, from its sensors and get its off-axis position $(\tilde{x}, \tilde{y})$ using motion capture.
2. Convert to off-axis velocities using
$$\begin{pmatrix} \dot{\tilde{x}} \\ \dot{\tilde{y}} \end{pmatrix} = \begin{pmatrix} \cos\theta & -l\sin\theta \\ \cos\theta & l\sin\theta \end{pmatrix} \begin{pmatrix} v \\ \omega \end{pmatrix}$$
from [4].
3. Calculate the desired next off-axis velocity for each robot **using the chosen algorithm**.
4. Convert this velocity back to (v, ω) using the inverse of step 2.
5. For simulations, the new heading and position is found by integration.
For physical robots, apply the new control velocity to each robot.

A3. Second-order Consensus

Fig. 3 and 4 were produced using:

$$\mathbf{u}_i = 5(\mathbf{g}_i + \mathbf{f}_i)$$

from [2], where the goal and formation errors respectively for robot i are:

$$\mathbf{g_i} = -\mathbf{x}_{ei} - \dot{\mathbf{x}}_i$$
$$\mathbf{f}_i = \sum_{j \in \mathcal{N}_i} (\mathbf{x}_{ej} - \mathbf{x}_{ei}) + (\dot{\mathbf{x}}_j - \dot{\mathbf{x}}_i)$$

and $\mathbf{x}_{ei} \stackrel{\text{def}}{=} \mathbf{x}_i - \mathbf{x}_i^d$, with off-axis position $\mathbf{x}_i$ and goal $\mathbf{x}_i^d$. $\mathcal{N}_i$ are the robots that robot i can communicate with.

- Since $\mathbf{u}_i$ is for acceleration control, get the next off-axis velocity for step 3 by integrating (i.e. in discrete time, $\dot{\mathbf{x}}_{i,new} = \dot{\mathbf{x}}_i + \mathbf{u}_i \Delta t$).
- $\mathbf{f}_i$ makes the robots approach their goals at the same rate. This is a weaker definition of formation-keeping than A4, which aims to preserve the shape of the initial formation.
- If some robots are initially much closer to their goal, the position term of $\mathbf{f}_i$ can make the robots move away from the goal to quickly reduce the formation error.

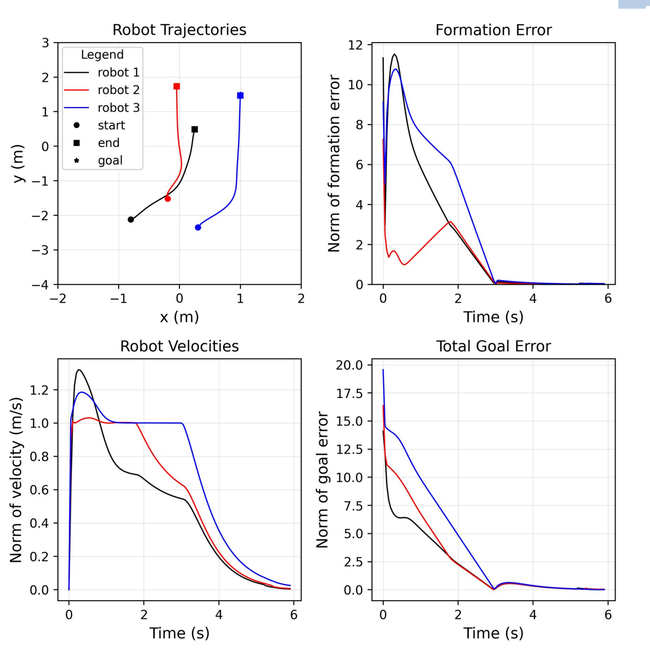


Fig 3. Second-order consensus simulation

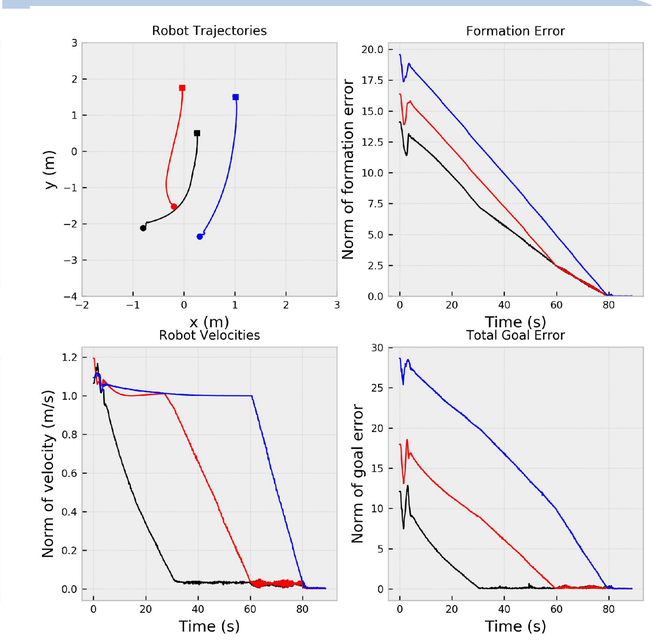
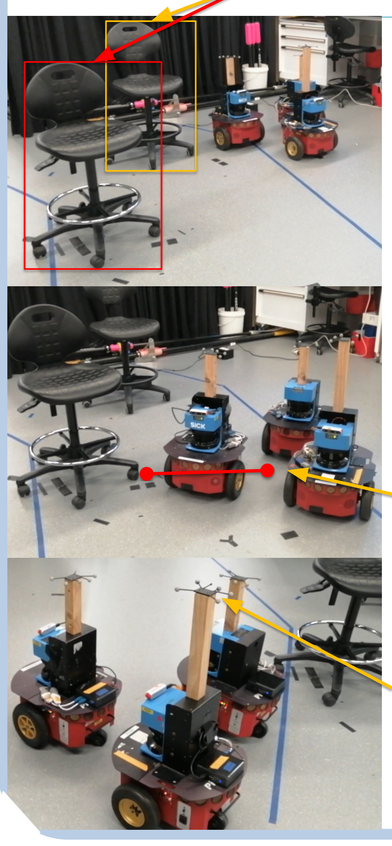


Fig 4. Second-order consensus experiment



Safety margin is relative to the centre of the formation

Motion capture is used since positions measured using odometry drift over time

A4. Obstacle Avoidance

Fig. 5 and 6 use the hierarchical quadratic program S2 in table 1 of [3]:

- This program puts the highest priority on minimising interdistance errors (formation keeping) whilst satisfying velocity limits.
- Out of these potential velocities, the second stage of the program finds a velocity that minimises the goal error under the constraint that the formation centre remains outside all safety margins.
- The experiment has larger interdistance errors than the simulation due to disturbances and numerical errors.
- In both cases, interdistances errors remain below 5cm, suitable for applications such as carrying a shared load.

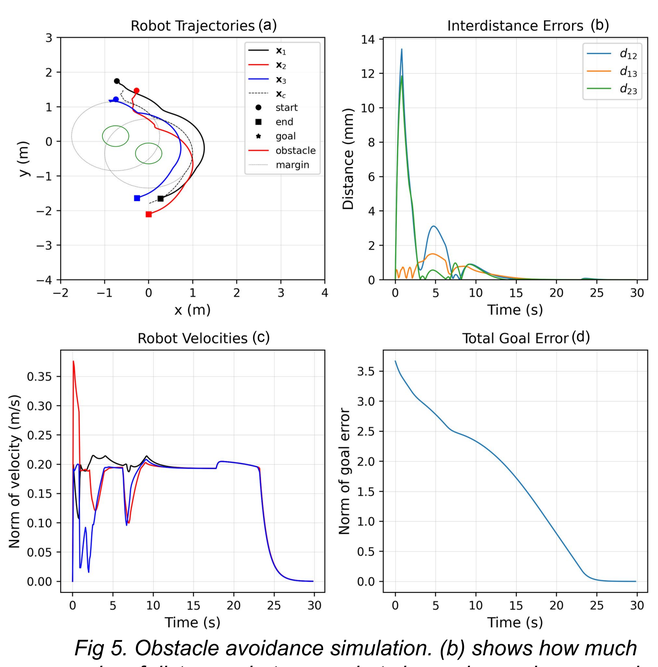


Fig 5. Obstacle avoidance simulation. (b) shows how much pairs of distances between robots have changed compared to the initial formation.

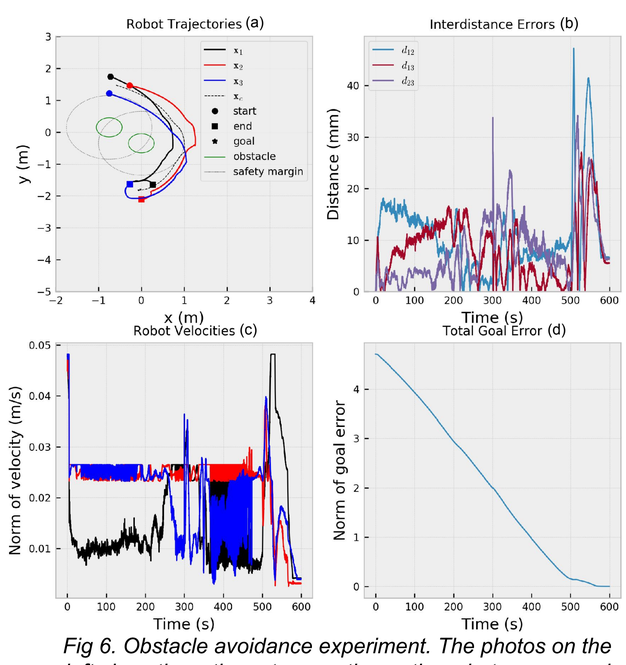


Fig 6. Obstacle avoidance experiment. The photos on the left show three time-steps as the as the robots go around the chairs marked by green circles in (a).

Conclusion

Summary

- Proposed several control laws for circular control of a single robot
- Proved the polar control law is globally asymptotically stable
- Recreated the simulation results in [1] and [2]
- Simulated and applied the algorithms in [2] and [3] to three wheeled robots

Future Work

A4 improves on A3 with obstacle avoidance and better formation keeping, but the following can be improved:

- Interdistance errors increase with sampling period. Investigate using numerical methods other than finite differences.
- A4 can converge to a local minima when safety margins overlap.
- Construct an algorithm which guarantees a worst-case interdistance error such as by making it a constraint.
- Investigate A4's robustness to moving obstacles.

References

[1] R. Olfati-Saber and R. M. Murray, "Consensus problems in networks of agents with switching topology and time-delays," *IEEE Transactions on Automatic Control*, vol. 49, no. 9, 2004.

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[3] D. Koung, O. Kermorgant, I. Fantoni, and L. Belouaer, "Cooperative multi-robot object transportation system based on hierarchical quadratic programming," *IEEE Robotics and Automation Letters*, vol. 6, no. 4, pp. 6466–6472, 2021.

[4] J. Cortes and M. Egerstedt, "Coordinated control of multi-robot systems: A survey," *SICE Journal of Control, Measurement, and System Integration*, vol. 10, no. 6, pp. 495–503, 2017.